

# ENEE 132 — Basic Electronics I (Semiconductor Devices)

## **Lecture 8 — Bipolar Junction Transistor**

21 July 2026

- **Part 1 — BJT Structure & Operation**

- The npn and pnp builds, forward-reverse bias, transistor action, and the three currents

- **Part 2 — BJT Characteristics and Parameters**

- $\beta_{DC}$  and  $\alpha_{DC}$ , the dc model, and solving a transistor bias circuit

- **Part 3 — Characteristic Curves & Operating Regions**

- The collector curves, cutoff, saturation, and the dc load line

- **Part 4 — Maximum Ratings & Wrap-Up**

- How  $\beta_{DC}$  varies, and the maximum ratings

# Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**

- Describe the npn and pnp BJT structure and the forward-reverse bias condition
- Explain transistor action and relate the three currents ( $I_E = I_C + I_B$ )
- Define and calculate  $\beta_{DC}$  and  $\alpha_{DC}$ . Solve a transistor bias circuit for its currents and voltages
- Interpret the collector characteristic curves and the cutoff, active, and saturation regions
- Apply the DC load line and the maximum transistor ratings

# Part 1 — BJT Structure & Operation

# From Diodes to Transistors

- **Where We've Been**

- The diode — one pn junction, a one-way valve for current flow

- **The Next Step**

- A BJT puts two pn junctions back-to-back to make a three-terminal device
- A small base current now controls a much larger collector current
- This is the basis of amplification and of electronic switching

# BJT Construction — The Three Regions

- **Three Doped Regions, Two Junctions**

- The BJT is constructed with three doped semiconductor regions separated by two pn junctions: the base-emitter (BE) and base-collector (BC) junctions
- npn = two n regions around a p region; pnp = two p regions around an n region

- **The Doping Is Deliberate**

- The emitter is heavily doped; the base is very thin and lightly doped; the collector is moderately doped
- “Bipolar” means both electrons and holes act as current carriers in the transistor
- A wire lead connects to each region: E, B, and C

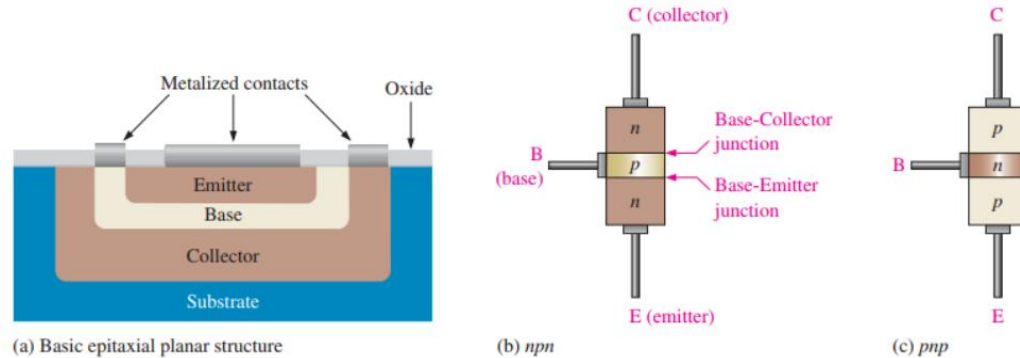


Fig 4.1 — Basic BJT construction: emitter, base, collector

# BJT Structure — npn and pnp

- **The Doping Is Deliberate**

On the schematic symbol, the arrow is on the emitter (it points out for npn, in for pnp)

The arrow points from P to N across the base-emitter junction — the direction of conventional emitter current when the transistor is on.

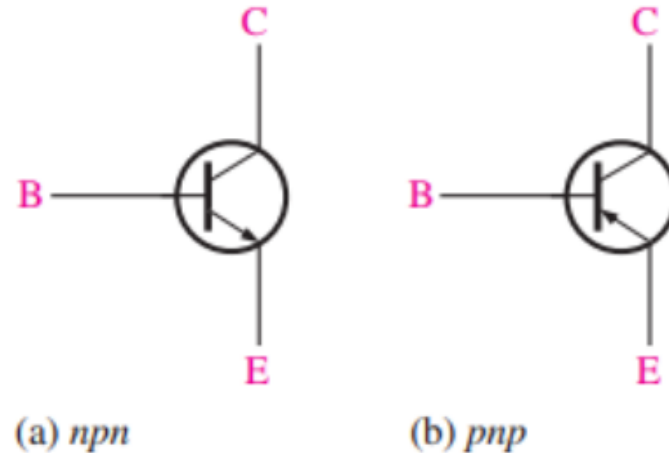


Fig 4.2 — Schematic symbols for npn and pnp BJTs

# Basic Operation — Forward-Reverse Bias

- **How a BJT Is Biased to Amplify**

- For normal (active) operation: the base-emitter junction is forward-biased and the base-collector junction is reverse-biased
- This condition is called forward-reverse bias

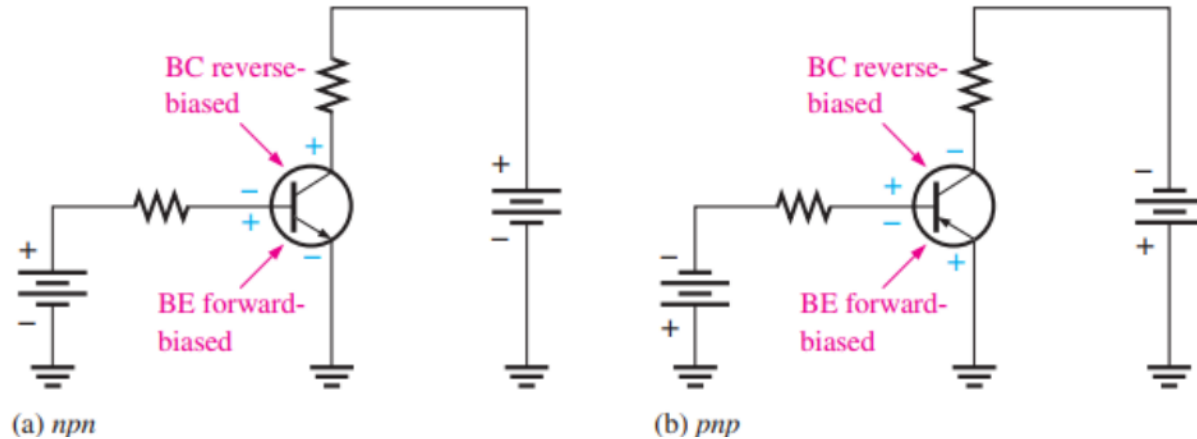


Fig 4.3 — Forward-reverse bias of a BJT



# Transistor Operation — How Current Flows

- **Inside the npn**

- The heavily doped emitter floods the forward-biased BE junction with free electrons, which diffuse into the thin, lightly doped base
- The base holds only a sparse population of holes (its majority carriers), so few of the arriving electrons find a hole to recombine with

- **The Key Result**

- The small fraction that do recombine leave through the metallic base lead as the base current
- Because the base is so thin, most electrons survive and are swept across the reverse-biased BC junction into the collector by the positive supply voltage
- These collector electrons return through the external circuit and back to the emitter along with the base current — so emitter current is slightly larger than collector current

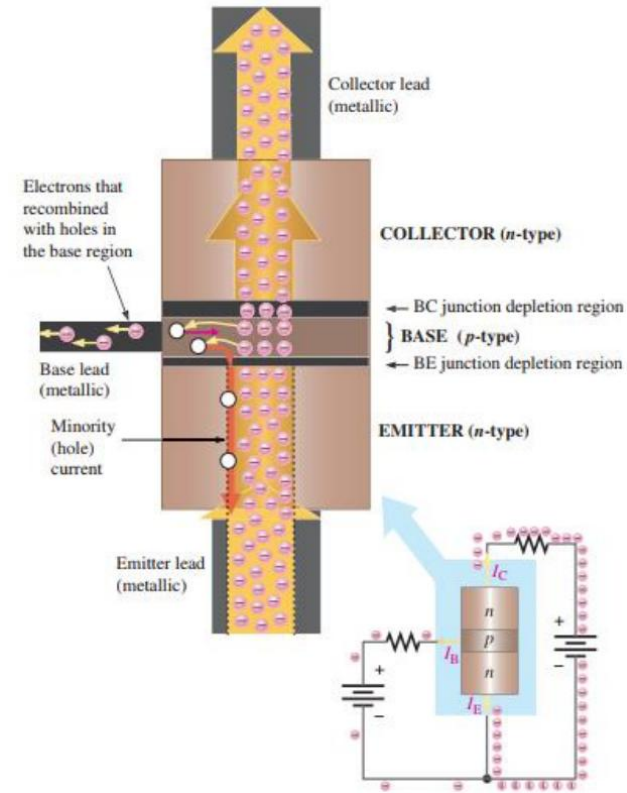


Fig 4.4 — BJT operation: electron flow in the npn

# The Three Transistor Currents

- **They Add Up**

- The emitter current is the sum of the collector and base currents:

$$I_E = I_C + I_B$$

- **Relative Sizes**

- $I_E$  is the largest;  $I_C$  is almost as large;  $I_B$  is very small by comparison
- The arrow on the emitter shows the direction of conventional current
- Capital-letter subscripts ( $I_C$ ,  $I_B$ ,  $I_E$ ) denote DC values

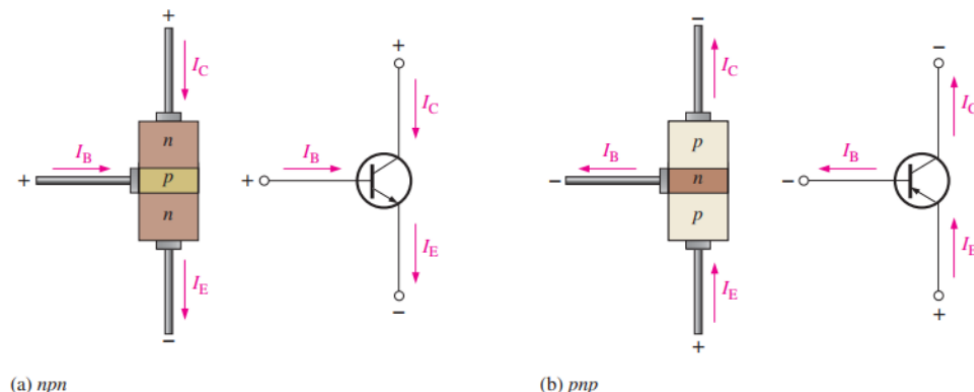


Fig 4.5 — The three transistor currents

## Quick Check #1 — Structure & Operation

- **Question 1**

- Name the three BJT terminals and the two junctions.

- **Question 2**

- For a BJT to operate as an amplifier, how is each junction biased?

- **Question 3**

- Why does only a small fraction of the emitter electrons become base current?

- **Question 4**

- Write the equation relating the three transistor currents.

- **Question 5**

- Which current is the largest, and which is the smallest?

# Part 2 — BJT CHARACTERISTICS AND PARAMETERS

# DC Beta and DC Alpha

## • DC Beta ( $\beta_{DC}$ ) — the Current Gain

- The dc current gain of a transistor is the ratio of dc collector current to the dc base current:

$$\beta_{DC} = \frac{I_C}{I_B}$$

- Typical values run from under 20 to 200 or more; on a transistor datasheet it's listed as equivalent hybrid parameter  $h_{FE}$

$$h_{FE} = \beta_{DC}$$

## • DC Alpha ( $\alpha_{DC}$ )

- The ratio of the dc collector current to the dc emitter current:

$$\alpha_{DC} = \frac{I_C}{I_E}$$

- Always slightly less than 1 (typically 0.95–0.99), because  $I_C$  is smaller than  $I_E$  by  $I_B$

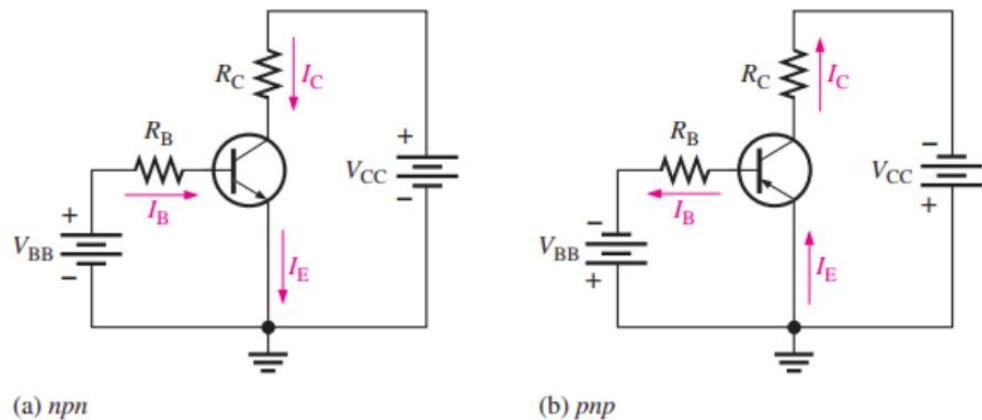


Fig 4.6 — Transistor dc bias circuits (npn and pnp)

## Worked Example — $\beta_{DC}$ and $\alpha_{DC}$

- **The Question (Example 4-1)**

- Determine the dc current gain  $\beta_{DC}$  and the emitter current  $I_E$  for a transistor where  $I_B = 50 \mu\text{A}$  and  $I_C = 3.65 \text{ mA}$

- **The Solution**

$$\beta_{DC} = \frac{I_C}{I_B} = \frac{3.65 \text{ mA}}{50 \mu\text{A}} = 73$$

$$I_E = I_C + I_B = 3.65 \text{ mA} + 50 \mu\text{A} = 3.70 \text{ mA}$$

# The Ideal DC Model of a BJT

- **Input Side — a Forward-Biased Diode**

- The base-emitter junction behaves like a forward-biased diode carrying the base current  $I_B$  (treat  $V_{BE} \approx 0.7\text{ V}$ )

- **Output Side — a Dependent Current Source**

- The collector behaves as a current source whose value depends on the base current:  $I_C = \beta_{DC} I_B$
- The diamond symbol marks a dependent source (an independent source uses a circle)
- This model holds in the active region — it is not valid in saturation

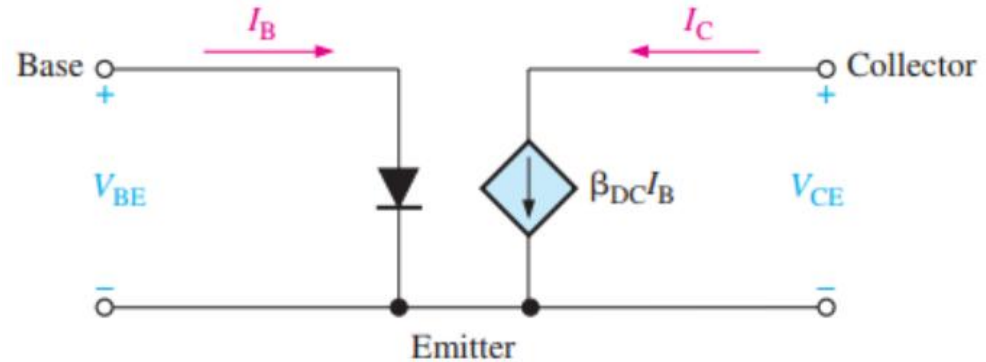


Fig 4.7 — Ideal dc model of an npn transistor

# The BJT Circuit Analysis

## • A Simple DC Model

- Input: the forward-biased BE junction carries the base current (treat  $V_{BE} \approx 0.7\text{ V}$ )
- Output: a dependent current source equal to  $\beta_{DC} I_B$

## • The Working Equations

$$V_{BE} \cong 0.7\text{ V}$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CB} = V_{CE} - V_{BE}$$

- Collector:  $I_C = \beta_{DC} I_B$
- Emitter:  $I_E = I_C + I_B$

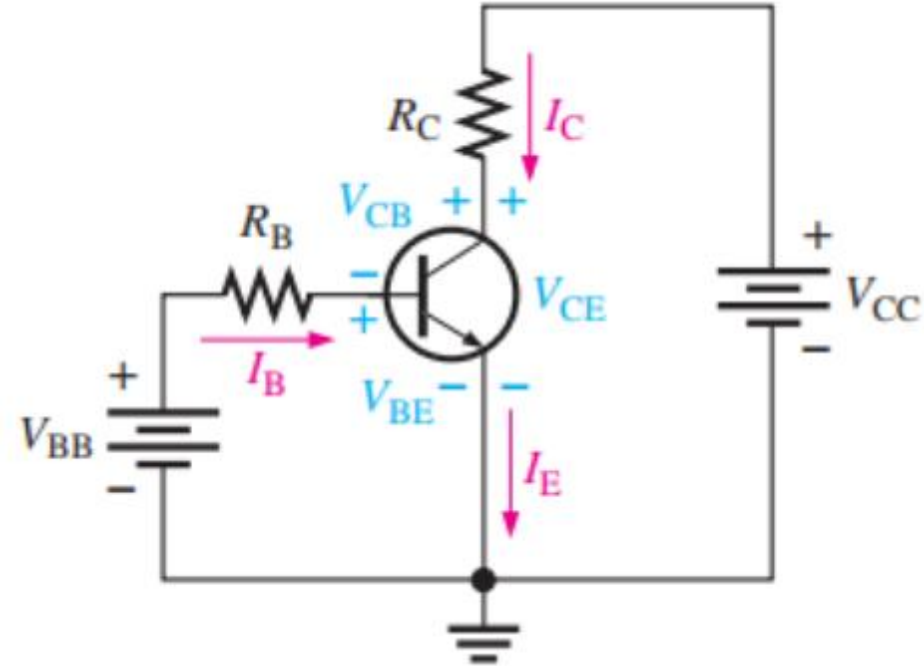


Fig 4.8 — Transistor currents and voltages



## Worked Example — Full DC Analysis

### • The Question (Example 4-2)

- Determine  $I_B$ ,  $I_C$ ,  $I_E$ ,  $V_{BE}$ ,  $V_{CE}$ , and  $V_{CB}$  in the circuit of Figure 4–9. The transistor has a  $\beta_{DC} = 150$ .

### • The Solution

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5\text{ V} - 0.7\text{ V}}{10\text{ k}\Omega} = 430\text{ }\mu\text{A}$$

$$I_C = \beta_{DC} I_B = (150)(430\text{ }\mu\text{A}) = 64.5\text{ mA}$$

$$I_E = I_C + I_B = 64.5\text{ mA} + 430\text{ }\mu\text{A} = 64.9\text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 10\text{ V} - (64.5\text{ mA})(100\text{ }\Omega) = 10\text{ V} - 6.45\text{ V} = 3.55\text{ V}$$

$$V_{CB} = V_{CE} - V_{BE} = 3.55\text{ V} - 0.7\text{ V} = 2.85\text{ V}$$

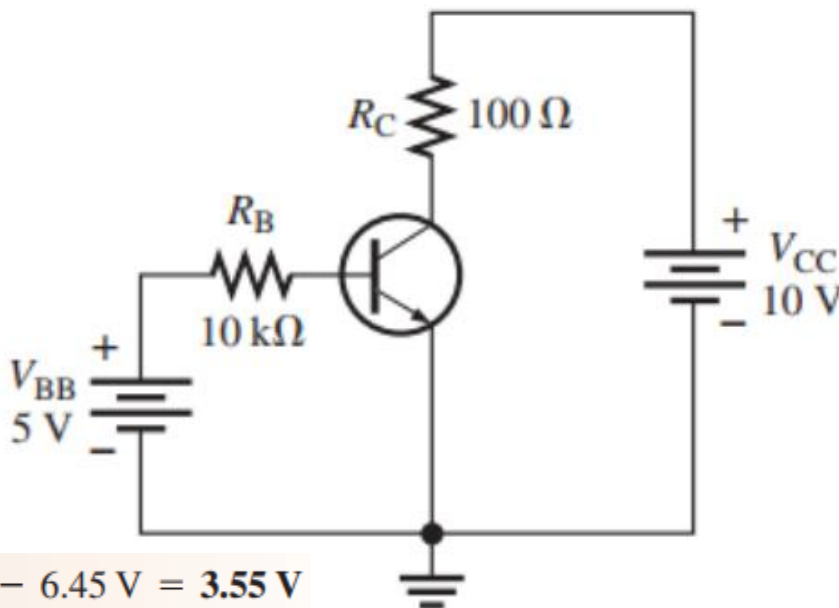


Fig 4.9 — Circuit for Example 4-2

## Quick Check #2 — Current Gain & DC Analysis

- **Question 1**

- Write the definitions of  $\beta_{DC}$  and  $\alpha_{DC}$ .

- **Question 2**

- Why is  $\alpha_{DC}$  always less than 1?

- **Question 3**

- If  $\beta_{DC} = 100$  and  $I_B = 40 \mu A$ , what is  $I_C$ ?

- **Question 4**

- What forward voltage do we assume across the base-emitter junction?

- **Question 5**

- Write the equation for  $V_{CE}$  in the basic bias circuit.

# Part 3 — Characteristic Curves & Operating Regions

# Collector Characteristic Curves I

## • Generating the Curves

- The circuit plots how collector current  $I_C$  varies with the collector-emitter voltage  $V_{CE}$  for specified values of base current  $I_B$
- Both  $V_{BB}$  and  $V_{CC}$  are variable sources:  $V_{BB}$  sets the base current;  $V_{CC}$  is swept to move  $V_{CE}$

## • The Starting Point

- With  $V_{BB}$  set for a chosen  $I_B$  and  $V_{CC}$  held at 0, the base sits at  $\approx 0.7$  V while both emitter and collector are at 0 V
- Both junctions are now forward-biased, so  $I_B$  drains to ground through the low-impedance BE path and  $I_C = 0$
- Two forward-biased junctions define the saturation region — where  $I_C$  reaches its maximum and no longer depends on  $I_B$

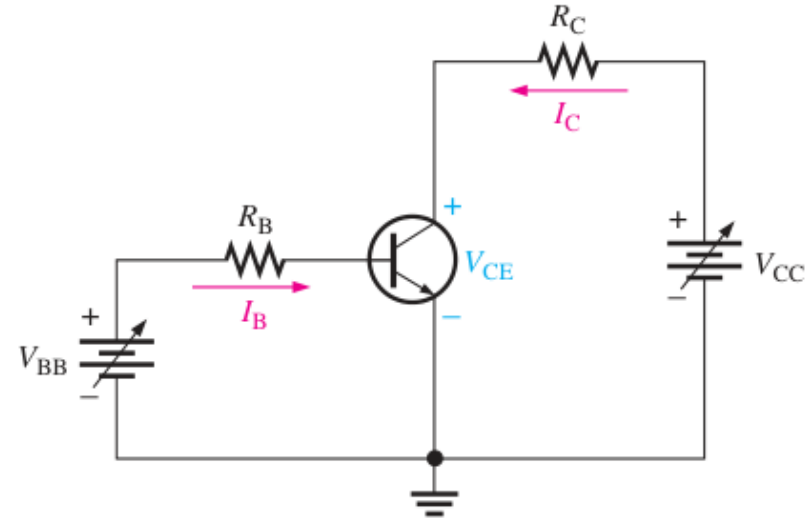


Fig 4.10 (a) — Collector characteristic curves - circuit

# Collector Characteristic Curves II

## • What They Show

- The collector characteristic curves show the relationship of the three transistor currents.
- The curve is shown for a fixed based current
- The first region is from  $A$  to  $B$  is the **saturation region**. As  $V_{CE}$  is increased,  $I_C$  increases until  $B$ .
- After reaching  $B$ , the curve flattens between points  $B$  and  $C$ , which is the **active region**.
- After  $C$ , is the **breakdown region**.

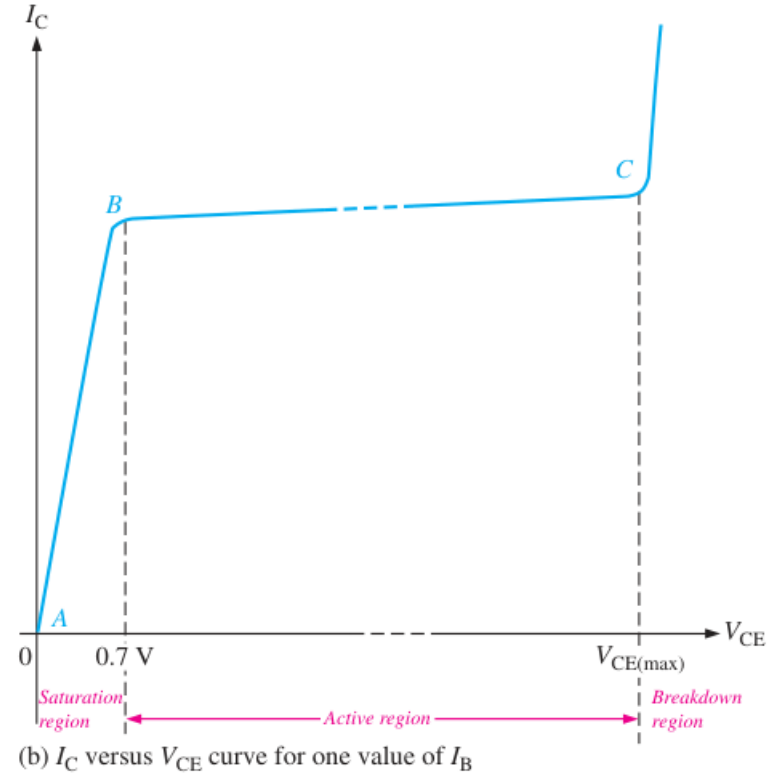


Fig 4.10 — Collector characteristic curves

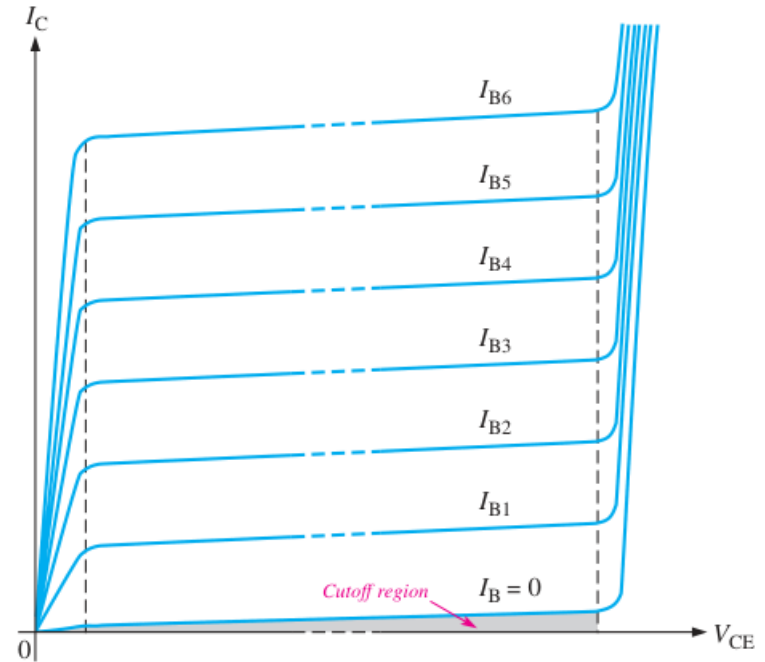
# Collector Characteristic Curves III

## • Family of Curves

- The collector characteristic curves illustrate the relationship of the three transistor currents.
- By setting up other values of base current, a family of collector curves is developed.

$$\beta_{DC} = \frac{I_C}{I_B}$$

- When  $I_B = 0$ , the transistor is in the cutoff region although there is a very small collector leakage current as indicated



(c) Family of  $I_C$  versus  $V_{CE}$  curves for several values of  $I_B$   
( $I_{B1} < I_{B2} < I_{B3}$ , etc.)

Fig 4.10 — Collector characteristic curves

# The Three Operating Regions

- **Saturation (low  $V_{CE}$ )**

- Both junctions forward-biased;  $I_C$  has risen to its maximum and no longer follows  $I_B$

- **Active / Linear (the useful region)**

- BE forward, BC reverse;  $I_C$  is essentially constant for a given  $I_B$ , set by  $I_C = \beta_{DC} \cdot I_B$
- This is where the transistor works as an amplifier

- **Breakdown (high  $V_{CE}$ )**

- The BC junction breaks down and  $I_C$  shoots up — a transistor must never be operated here

## Example 4-3 — The Test Circuit

### • Question

- Sketch an ideal family of collector curves for the circuit in Figure 4–11 for  $I_B$  5  $\mu\text{A}$  to 25  $\mu\text{A}$  in 5  $\mu\text{A}$  increments. Assume  $\beta_{DC} = 100$  and that  $V_{CE}$  does not exceed breakdown.

### • What We Compute

- In the active region the collector current is  $I_C = \beta_{DC} \cdot I_B$  for each value of  $I_B$
- Plotting  $I_C$  versus  $V_{CE}$  for each  $I_B$  traces one curve in the family (next slide)

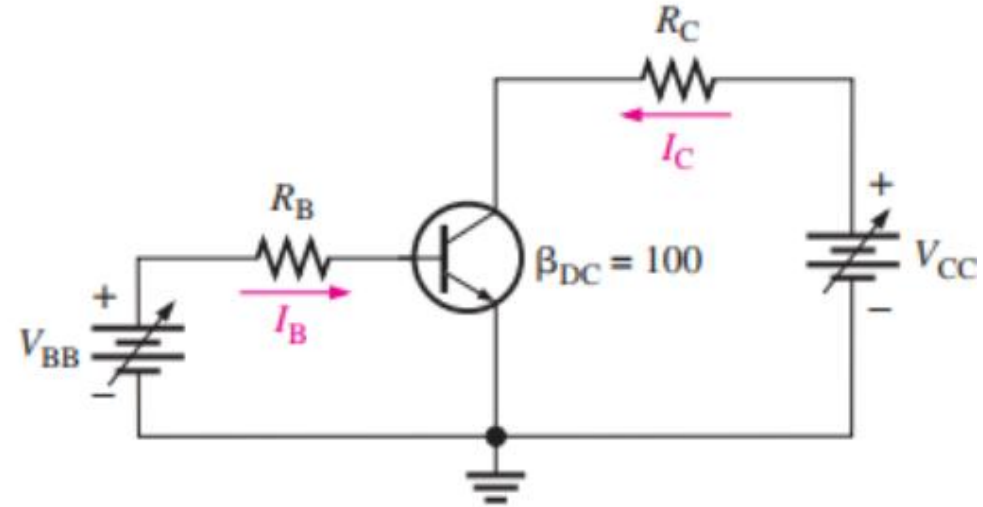


Fig 4.11 — Circuit for Example 4-3



# Worked Example — A Family of Curves

- Solution**

| $I_B$             | $I_C$            |
|-------------------|------------------|
| $5\ \mu\text{A}$  | $0.5\ \text{mA}$ |
| $10\ \mu\text{A}$ | $1.0\ \text{mA}$ |
| $15\ \mu\text{A}$ | $1.5\ \text{mA}$ |
| $20\ \mu\text{A}$ | $2.0\ \text{mA}$ |
| $25\ \mu\text{A}$ | $2.5\ \text{mA}$ |

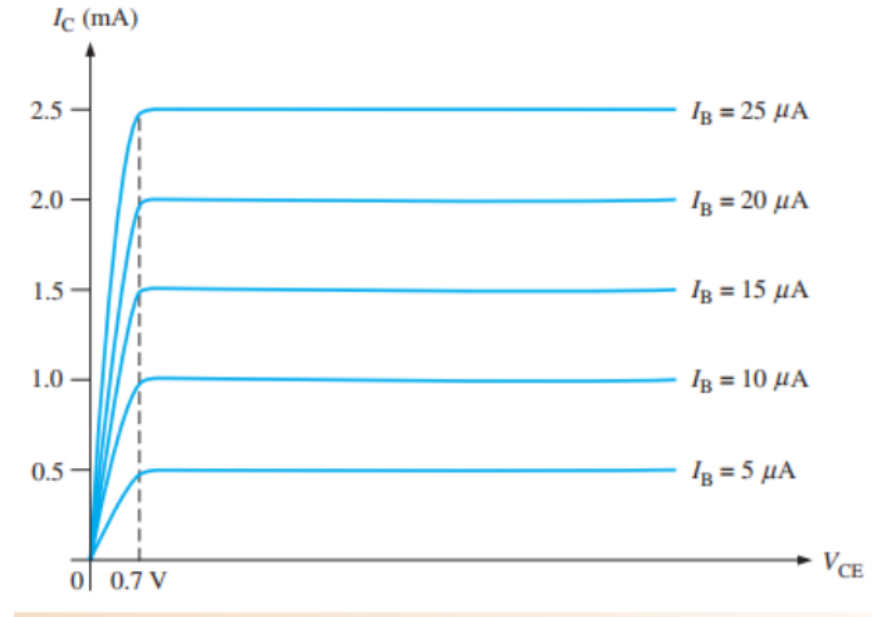


Fig 4.12 — Family of collector curves (Example 4-3)

# Cutoff — The Transistor Off

- **No Base Current**

- In a BJT, **cutoff** is the condition in which there is no base current, which results in only an extremely small leakage current ( $I_{CEO}$ ) in the collector circuit.
- When  $I_B = 0$  (base open), the transistor is in cutoff — its non-conducting state
- Both junctions are reverse-biased

- **What's Left**

- Only a very small amount of collector leakage current,  $I_{CEO}$ , flows — small enough to neglect
- With  $I_C \approx 0$  there's no drop across  $R_C$ , so  $V_{CE} = V_{CC}$
- The subscript CEO represents collector-to-emitter with the base open

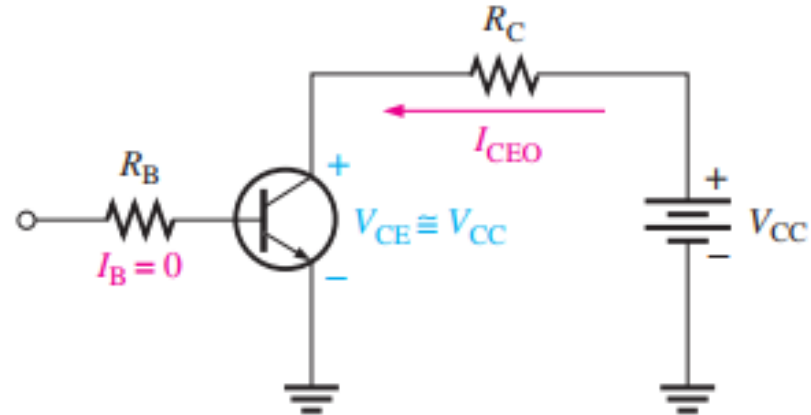


Fig 4.13 — Cutoff: base open, only collector leakage current flows

# Saturation — The Transistor Fully On

## • Driving $I_B$ Up

- In a BJT, **saturation** is the condition in which there is maximum collector current. The saturation current is determined by the external circuit ( $V_{CC}$  and  $R_C$  in this case) because the collector-emitter voltage is minimum ( $\approx 0.2$  V)
- As  $I_B$  increases,  $I_C$  increases and  $V_{CE}$  falls ( $V_{CE} = V_{CC} - I_C R_C$ )
- When  $V_{CE}$  drops to  $V_{CE}(\text{sat})$ , the BC junction becomes forward-biased and  $I_C$  can rise no further

## • The Saturation Current

- $I_C(\text{sat}) = (V_{CC} - V_{CE}(\text{sat})) / R_C$
- In saturation, an increase of base current has no effect on the collector circuit and the relation  $I_C = \beta_{DC} I_B$  is no longer valid.

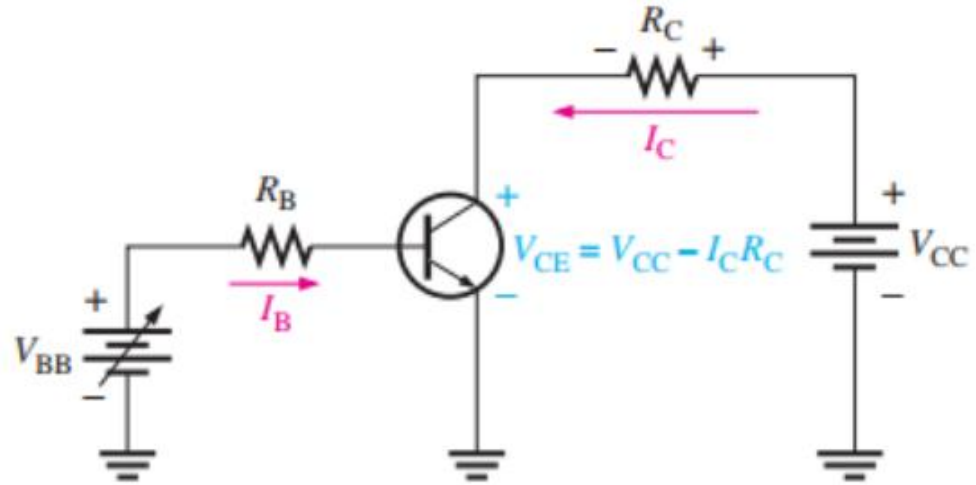


Fig 4.14 — Saturation:  $V_{CE} = V_{CC} - I_C R_C$

# The DC Load Line

- **A Line Across the Curves**

- Drawn on the family of curves, it connects the two extreme operating points
- Bottom (cutoff):  $I_C = 0$  and  $V_{CE} = V_{CC}$
- Top (saturation):  $I_C = I_{C(sat)}$  and  $V_{CE} = V_{CE(sat)}$

- **In Between**

- Every point along the line in the active region is a possible operating (Q) point

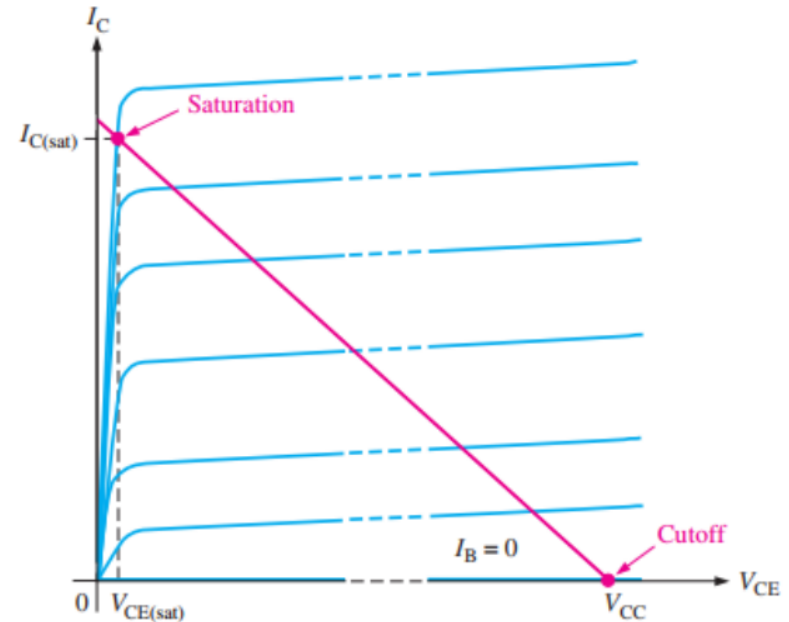


Fig 4.15 — DC load line on the family of curves

# Worked Example — Is It Saturated?

## • The Question (Example 4-4)

- Determine whether or not the transistor in Figure 4–16 is in saturation. Assume  $V_{CE(sat)} = 0.2 \text{ V}$ .

## • The Solution — Compare

- First, determine  $I_{C(sat)}$ .

$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C} = \frac{10 \text{ V} - 0.2 \text{ V}}{1.0 \text{ k}\Omega} = \frac{9.8 \text{ V}}{1.0 \text{ k}\Omega} = 9.8 \text{ mA}$$

- Now, see if  $I_B$  is large enough to produce  $I_{C(sat)}$ .

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{3 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = \frac{2.3 \text{ V}}{10 \text{ k}\Omega} = 0.23 \text{ mA}$$

$$I_C = \beta_{DC} I_B = (50)(0.23 \text{ mA}) = 11.5 \text{ mA}$$

- 11.5 mA exceeds 9.8 mA, so the transistor is saturated — actual  $I_C$  stays at 9.8 mA

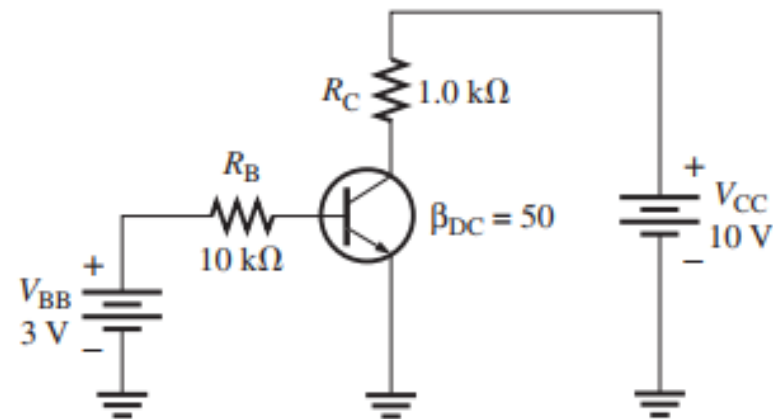


Fig 4.16 — Circuit for Example 4-4

## Quick Check #3 — Curves & Regions

- **Question 1**

- What do the collector characteristic curves plot, and what is held fixed for each curve?

- **Question 2**

- In which region does the transistor act as an amplifier, and how are the junctions biased there?

- **Question 3**

- In cutoff, what is  $V_{CE}$ , and why?

- **Question 4**

- Write the equation for  $I_{C(sat)}$ .

- **Question 5**

- What two points does the dc load line connect?

# Part 4 — Maximum Ratings & Wrap-Up

# $\beta_{DC}$ Is Not Constant

- **It Drifts with Temperature**

- $\beta_{DC}$  (hFE) varies with collector current and with temperature — it is not a fixed number
- At fixed temperature, raising  $I_C$  pushes  $\beta_{DC}$  up to a peak, then back down; at fixed  $I_C$ ,  $\beta_{DC}$  rises with temperature

- **And It Varies Between Devices**

- Even at the same  $I_C$  and temperature,  $\beta_{DC}$  differs unit-to-unit for the same part number
- Datasheets usually quote a minimum value,  $\beta_{DC(min)}$ , at a specified  $I_C$

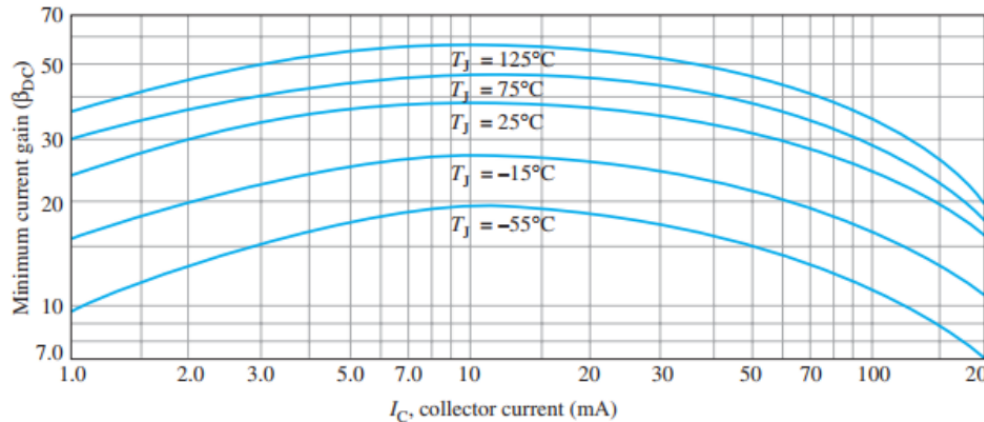


Fig 4.17 — Variation of beta-DC with  $I_C$  and temperature



# Maximum Transistor Ratings

## • The Limits

- Datasheets specify maximums for  $V_{CB}$ ,  $V_{CE}$ ,  $V_{EB}$ ,  $I_C$ , and power dissipation  $P_{D(max)}$

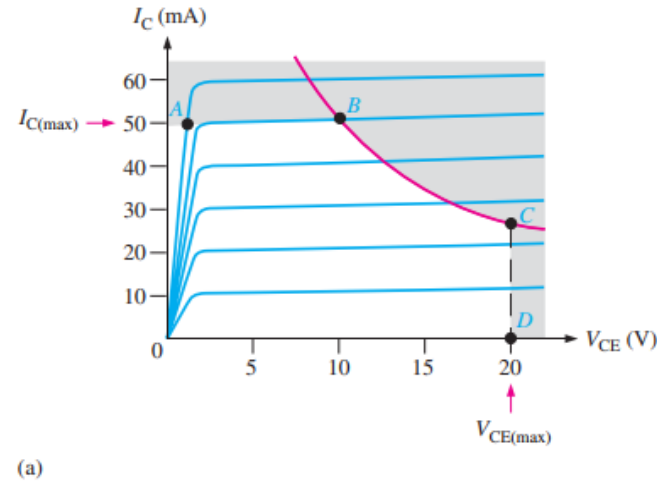
## • The Power Constraint

- The product of  $V_{CE}$  and  $I_C$  must not exceed the maximum power dissipation. Both  $V_{CE}$  and  $I_C$  cannot be maximum at the same time. If  $V_{CE}$  is maximum,  $I_C$  can be calculated as

$$I_C = \frac{P_{D(max)}}{V_{CE}}$$

- If  $I_C$  is maximum,  $V_{CE}$  can be calculated by rearranging the previous equation as follows:

$$V_{CE} = \frac{P_{D(max)}}{I_C}$$



| $P_{D(max)}$ | $V_{CE}$ | $I_C$  |
|--------------|----------|--------|
| 500 mW       | 5 V      | 100 mA |
| 500 mW       | 10 V     | 50 mA  |
| 500 mW       | 15 V     | 33 mA  |
| 500 mW       | 20 V     | 25 mA  |

Fig 4.18 — Maximum power dissipation curve and ratings

## Worked Example — Maximum Collector Current

- **The Question (Example 4-5)**

- A certain transistor is to be operated with  $V_{CE} = 6\text{ V}$ . If its maximum power rating is 250 mW, what is the most collector current that it can handle?

- **The Solution**

$$I_C = \frac{P_{D(\max)}}{V_{CE}} = \frac{250\text{ mW}}{6\text{ V}} = 41.7\text{ mA}$$

- **The Takeaway**

- This is the maximum current for this particular value of  $V_{CE}$ . The transistor can handle more collector current if  $V_{CE}$  is reduced, as long as  $P_{D(\max)}$  and  $I_{C(\max)}$  are not exceeded.

## Worked Example — Maximum Supply Voltage

### • The Question (Example 4-6)

- The transistor in Figure 4–19 has the following maximum ratings:  $P_{D(max)} = 800 \text{ mW}$ ,  $V_{CE(max)} = 15 \text{ V}$ , and  $I_{C(max)} = 100 \text{ mA}$ . Determine the maximum value to which  $V_{CC}$  can be adjusted without exceeding a rating. Which rating would be exceeded first?

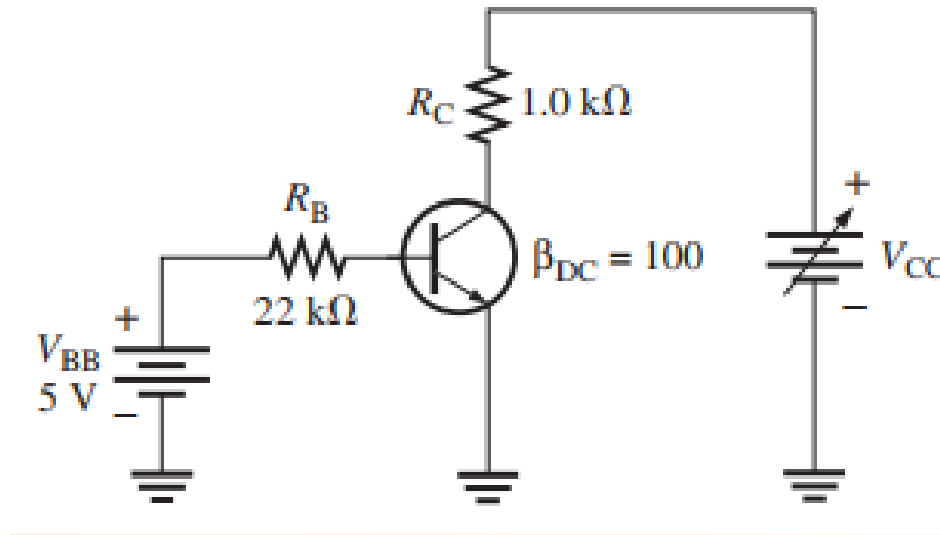


Fig 4.19 — Circuit for Example 4-6

## Worked Example — Maximum Supply Voltage

First, find  $I_B$  so that you can determine  $I_C$ .

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5\text{ V} - 0.7\text{ V}}{22\text{ k}\Omega} = 195\text{ }\mu\text{A}$$

$$I_C = \beta_{DC} I_B = (100)(195\text{ }\mu\text{A}) = 19.5\text{ mA}$$

$I_C$  is much less than  $I_{C(\max)}$  and ideally will not change with  $V_{CC}$ . It is determined only by  $I_B$  and  $\beta_{DC}$ .

The voltage drop across  $R_C$  is

$$V_{R_C} = I_C R_C = (19.5\text{ mA})(1.0\text{ k}\Omega) = 19.5\text{ V}$$

Now you can determine the value of  $V_{CC}$  when  $V_{CE} = V_{CE(\max)} = 15\text{ V}$ .

$$V_{R_C} = V_{CC} - V_{CE}$$

So,

$$V_{CC(\max)} = V_{CE(\max)} + (V_{R_C} = 15\text{ V} + 19.5\text{ V} = \mathbf{34.5\text{ V}})$$

$V_{CC}$  can be increased to 34.5 V, under the existing conditions, before  $V_{CE(\max)}$  is exceeded. However, at this point it is not known whether or not  $P_{D(\max)}$  has been exceeded.

$$P_D = V_{CE(\max)} I_C = (15\text{ V})(19.5\text{ mA}) = 293\text{ mW}$$

Since  $P_{D(\max)}$  is 800 mW, it is *not* exceeded when  $V_{CC} = 34.5\text{ V}$ . So,  $V_{CE(\max)} = 15\text{ V}$  is the limiting rating in this case. If the base current is removed causing the transistor to turn off,  $V_{CE(\max)}$  **will be exceeded first** because the entire supply voltage,  $V_{CC}$ , will be dropped across the transistor.

# Derating Power Dissipation

- **Why the Power Rating Shrinks with Heat**

- $P_{D(max)}$  is usually specified at 25 °C; at higher temperatures the transistor can safely dissipate less
- Datasheets give a derating factor (e.g. in mW per °C) to find  $P_{D(max)}$  above 25 °C

- **Worked Example (Example 4-7)**

- A certain transistor has a  $P_{D(max)}$  of 1 W at 25°C. The derating factor is 5 mW/°C. What is the  $P_{D(max)}$  at a temperature of 70°C?

- **Solution**

- The change (reduction) in  $P_{D(max)}$  is

$$\Delta P_{D(max)} = (5 \text{ mW/°C})(70^\circ\text{C} - 25^\circ\text{C}) = (5 \text{ mW/°C})(45^\circ\text{C}) = 225 \text{ mW}$$

- Therefore, the  $P_{D(max)}$  at 70°C is

$$1 \text{ W} - 225 \text{ mW} = 775 \text{ mW}$$

## Quick Check #4 — Maximum Ratings

- **Question 1**

- Name three maximum ratings given on a BJT datasheet.

- **Question 2**

- Why can't  $V_{CE}$  and  $I_C$  both be at their maximum at the same time?

- **Question 3**

- At  $V_{CE} = 5\text{ V}$  with  $P_{D(\max)} = 500\text{ mW}$ , what is the maximum  $I_C$ ?

# Key Takeaways from Today

- A BJT is two pn junctions back-to-back (nnp or pnp) with three regions: emitter, base, collector
- For amplification: BE junction forward-biased, BC junction reverse-biased (forward-reverse bias)
- The currents add:  $I_E = I_C + I_B$ , with  $I_B$  the smallest
- Current gain:  $\beta_{DC} = I_C/I_B$ ; also  $\alpha_{DC} = I_C/I_E < 1$
- DC analysis:  $I_B = (V_{BB} - V_{BE})/R_B$ ,  $I_C = \beta_{DC} \cdot I_B$ ,  $V_{CE} = V_{CC} - I_C \cdot R_C$  (with  $V_{BE} \approx 0.7 \text{ V}$ )
- Three regions: cutoff ( $I_B = 0$ ,  $V_{CE} = V_{CC}$ ), active (amplifier,  $I_C = \beta_{DC} \cdot I_B$ ), saturation ( $I_C(\text{sat}) = (V_{CC} - V_{CE}(\text{sat}))/R_C$ )
- Respect the maximum ratings:  $V_{CE} \times I_C \leq PD(\text{max})$ , and derate at higher temperatures

## Coming Up — Week 9: BJT as Amplifier & as Switch

- **Next Topic — The BJT in Action (Lecture Notes pp. 136–144)**
  - Using the transistor in the active region to amplify a signal (voltage gain)
  - Driving it between cutoff and saturation to act as a switch
  - Tying together the load line, the Q-point, and the operating regions from today
- **To Prepare**
  - Make sure today's dc analysis and the three regions are solid
- **Reminder**
  - Read pp. 136–144 before class



Thank You — Questions?